

Sharp gaps for positive multilinear relaxations

Abstract

Luedtke, Namazifar, and Linderoth conjectured that for multilinear functions with positive coefficients on nonnegative boxes, the termwise gap (the sum of the envelope widths of the individual terms) is at most a universal constant times the convex-hull gap. We disprove this conjecture with a sparse family of unit-coefficient polynomials on unit cubes. At a strictly interior point, the L th member has termwise gap L and exact hull gap $s + (L - s)2^{-s}$, where $s = \log_2 L + O(1)$ is an explicit cutoff; the ratio therefore grows like $L/\log_2 L$. A finite inequality bounds the hull gap and an explicit probability law attains the bound. We then determine the sharp growth of the worst-case ratio over all finite nonnegative boxes: $R(d) \sim \ln d / \ln \ln d$ in the maximum degree d and $C(n) \sim \ln n / \ln \ln n$ in the number n of variables. The upper bound uses one mixture of three probability laws that bounds every term at once and applies to the original terms on every finite nonnegative box. An accompanying standalone Lean project verifies the disproof, the exact hull-gap formula, the explicit finite upper bound, and both asymptotic limits.

1 Introduction

Global solvers for nonconvex problems usually relax each nonlinear term separately. For a multilinear function, the termwise relaxation replaces each monomial by its own convex and concave envelopes; for bilinear terms these are the McCormick inequalities. The convex hull of the whole graph is the strongest relaxation, but its standard description uses all vertices of the box, exponentially many [2]. This paper asks how much the termwise relaxation can lose.

Setting. Let $B = \prod_{i=1}^n [\ell_i, u_i]$ be a finite box with $0 \leq \ell_i \leq u_i$, and consider

$$f(x) = \sum_{e \in E} a_e m_e(x), \quad m_e(x) = \prod_{i \in e} x_i, \quad a_e \geq 0, \quad (1)$$

where E is a set of distinct subsets of $[n]$. Zero coefficients can be deleted, so this class contains the positive-coefficient class. Empty and singleton supports are allowed and contribute only affine terms. For a continuous function g on B , write $\text{vex}_B g$ for its greatest convex underestimator and $\text{cav}_B g$ for its least concave overestimator. The *hull gap* and the *termwise gap* are

$$H_B(f; x) = \text{cav}_B f(x) - \text{vex}_B f(x), \quad (2)$$

$$T_B(f; x) = \sum_{e \in E} [\text{cav}_B(a_e m_e)(x) - \text{vex}_B(a_e m_e)(x)]. \quad (3)$$

The hull gap is the vertical width of the convex hull of the graph of f ; the termwise gap gives each original monomial its exact graph hull separately. Always $0 \leq H_B \leq T_B$, and we form T_B/H_B only when $H_B > 0$. For $d \geq 2$, let $R(d)$ be the supremum of T_B/H_B over polynomials (1) of maximum degree at most d , in any finite dimension, over all such boxes and all points with positive hull gap. Let $C(n)$ be the analogous supremum over exactly n ambient coordinates, with no separate degree restriction. Unused coordinates, boundary points, and fixed coordinates ($\ell_i = u_i$) are permitted in both definitions.

Prior work. Luedtke et al. [2] studied this comparison. With positive coefficients on a nonnegative box, the concave envelope of f is the sum of the concave envelopes of its terms, so all of the loss is on the convex side. For bilinear functions with positive coefficients on the unit cube, they proved that the McCormick gap, which equals the termwise gap for bilinear terms, is less than twice the hull gap, with a factor that depends on the chromatic number of the graph of the terms. Numerical experiments then led to their Conjecture 4.1 [2, p. 349]: for positive multilinear functions on nonnegative boxes, T_B/H_B is bounded by a universal constant. The constant is meant to be uniform in both degree and dimension. Without the sign condition, even bilinear functions behave differently: Boland et al. [1] showed that on the unit cube the ratio of the McCormick gap to the hull gap can grow like $\sqrt{n}$ and is always $O(\sqrt{n})$. We do not claim an exhaustive literature search for priority.

Results.

- *Disproof* (Theorem 5). The family f_L of (5) has unit coefficients on a unit cube and degree $2^{L-1} + 1$. At a strictly interior point, $T_L = L$ (Lemma 3) while $H_L \leq \lceil \log_2 L \rceil + 1$ (Lemma 4 with $s = \lceil \log_2 L \rceil$), so T/H is unbounded.
- *Exact gap* (Theorem 6). For an explicit cutoff $s = \log_2 L + O(1)$, $H_L = s + (L - s)2^{-s}$ exactly, and an explicit law attains it. Thus $T_L/H_L \sim L/\log_2 L$ (16).
- *Sharp growth* (Theorem 7). $R(d) \sim \ln d / \ln \ln d$ and $C(n) \sim \ln n / \ln \ln n$, with leading constant one. The upper half is an explicit finite bound $T_B \leq ZH_B$ for every degree d (Theorem 8), valid for the original terms on every finite nonnegative box. The lower half comes from the family above.
- *Formal verification* (Appendix B) of all of these results in Lean.

Key idea. By Lemma 1, $\text{vex}_B f(x)$ is the minimum of $\mathbb{E}f(X)$ over probability laws on the box vertices with mean x . In the termwise gap each term may use its own minimizing law, while the hull gap requires one law for all terms together. The family is built so that the individual minimizing laws conflict. For the upper bound we construct one law, depending only on the variable means and the degree, that captures at least a fraction $1/Z$ of every term's gap at once.

Scope. Our upper bounds concern the exact termwise envelopes in (3), not recursive McCormick relaxations of higher-degree products, which can be weaker on general boxes [2]. A pointwise width ratio also does not by itself give an approximation ratio for a constrained optimization problem or an algorithm for evaluating a convex envelope.

Section 2 gives the vertex-law description of graph hulls; Sections 3–5 and Appendix A prove the results in the order listed.

2 Graph hulls and probability laws

We omit the subscript B on a unit cube. A polynomial is multilinear when it is affine in each coordinate separately.

Lemma 1 (Vertex laws). *For a multilinear g on a finite box B , let $\mathcal{P}_B(x)$ be the probability laws on its vertices having mean x . Then*

$$\text{vex}_B g(x) = \min_{P \in \mathcal{P}_B(x)} \mathbb{E}_P g(X), \quad \text{cav}_B g(x) = \max_{P \in \mathcal{P}_B(x)} \mathbb{E}_P g(X).$$

Both extrema are attained, and these are the endpoints of the vertical slice of the original continuous graph hull.

Proof. Round each nonfixed coordinate of a point $z \in B$ independently to its two endpoints, preserving its mean. Repeated use of separate affinity gives $\mathbb{E}X = z$ and $\mathbb{E}g(X) = g(z)$. A fixed coordinate stays fixed. Thus every continuous graph point lies in the convex hull of the vertex graph. The reverse hull inclusion is immediate. A convex combination of vertex graph points is exactly a vertex law with its mean and expected function value. Laws with a specified mean form a nonempty compact polytope; hence minimizing and maximizing the expected value attain the two endpoints. $\square$

Lemma 2 (Individual envelopes and a common upper law). *For a nonempty support e on the unit cube,*

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max \left(0, \sum_{i \in e} x_i - |e| + 1 \right). \quad (4)$$

A single law attains the upper envelope of every monomial simultaneously. Consequently, $\text{cav } f(x) = \sum_e a_e \text{cav } m_e(x)$ when $a_e \geq 0$.

Proof. On a binary vertex, a monomial indicates simultaneous success of all its coordinates. The joint-success probability is at most the smallest success marginal and at least one minus the sum of the failure marginals, or zero if that is larger. For the upper bound, take one uniform $U \in [0, 1]$ and set $X_i = \mathbf{1}\{U \leq x_i\}$ for every coordinate. This attains all upper bounds at once.

For the lower bound for one support, place intervals of lengths $1 - x_i$ consecutively on a circle of circumference one. Their union has length $\min(1, \sum_{i \in e} (1 - x_i))$: before one full turn the intervals are disjoint up to endpoints, and after a full turn they cover the circle. Let each interval be the corresponding failure event. This realizes the claimed joint-success probability. Sample unused coordinates with their prescribed means to complete the law. Nonnegative coefficients allow the common upper equalities to be summed. A constant term is immediate. $\square$

These elementary envelope facts are also part of the classical multilinear relaxation theory discussed in Luedtke et al. [2].

3 A sparse family with unbounded ratio

Fix an integer $L \geq 2$ and put $m = 2^L$. Use leaves $z_0, \dots, z_{m-1}$ and anchors $a_1, \dots, a_L$. At level j , partition the leaf indices into consecutive blocks

$$\mathcal{B}_j = \{ \{b2^{L-j}, \dots, (b+1)2^{L-j} - 1\} : 0 \leq b < 2^j \}.$$

The partitions are nested. Define the polynomial and evaluation point by

$$f_L(a, z) = \sum_{j=1}^L \sum_{B \in \mathcal{B}_j} a_j \prod_{i \in B} z_i, \quad a_j = 2^{-j}, \quad z_i = 1 - 2^{-L}. \quad (5)$$

We write T_L and H_L for its two gaps at this point. Each support is distinguished by its anchor and its block, so all coefficients are one on distinct squarefree supports. The number of variables, number of monomials, maximum degree, and number of variable occurrences are

$$n_L = 2^L + L, \quad |E_L| = 2^{L+1} - 2, \quad d_L = 2^{L-1} + 1, \quad \sum_{e \in E_L} |e| = L2^L + 2^{L+1} - 2. \quad (6)$$

The counts follow by summing 2^j terms of size $2^{L-j} + 1$ at each level. Every prescribed coordinate lies strictly between zero and one.

Lemma 3 (Termwise gap and failure representation). *For the family (5), $T_L = L$, the upper envelope is L , and $H_L > 0$. Under any binary law with the prescribed means, let A_j be the anchor, R the number of failed leaves, and N_j the number of level- j blocks containing a failed leaf. Then*

$$\mathbb{E}R = 1, \quad \mathbb{E}A_j = 2^{-j}, \quad H_L = \max_P \mathbb{E}_P \sum_{j=1}^L A_j N_j, \quad N_j \leq \min(2^j, R). \quad (7)$$

The maximum ranges over all binary laws with all original coordinate means, without independence assumptions.

Proof. At level j , a support has $k_j = 2^{L-j}$ leaves and one anchor. Its lower envelope is $\max(0, 2^{-j} - k_j 2^{-L}) = 0$, and its upper envelope is 2^{-j} . There are 2^j such terms. Lemma 2 therefore gives $T_L = \text{cav } f_L = L$. The value at the original point is

$$f_L(a, z) = \sum_{j=1}^L (1 - 2^{-L}) 2^{L-j} < L.$$

Since $\text{vex } f_L \leq f_L$, the hull gap is positive.

The expected number of failures is $m2^{-L} = 1$. At every binary vertex, $f_L = \sum_j A_j(2^j - N_j)$. Taking expectations gives $\mathbb{E}f_L = L - \mathbb{E} \sum_j A_j N_j$. Apply Lemma 1 to minimize $\mathbb{E}f_L$. Finally, the blocks of each partition are disjoint, so each hit block needs a different failed leaf. $\square$

The next bound holds for every law in (7).

Lemma 4 (A finite cap certificate). *For $r \geq 0$, $A_j \in \{0, 1\}$, and integers $0 \leq s \leq L$,*

$$\sum_{j=1}^L A_j \min(2^j, r) \leq sr + \sum_{j=s+1}^L A_j 2^{j-s}. \quad (8)$$

Consequently,

$$H_L \leq s + (L - s)2^{-s} \quad (0 \leq s \leq L). \quad (9)$$

Proof. Put $c_j = \min(2^j, r)$. For $j > s$,

$$A_j c_j \leq c_j - c_{j-s} + A_j 2^{j-s}.$$

If $A_j = 0$ this uses $c_j \geq c_{j-s}$; if $A_j = 1$ it uses $c_{j-s} \leq 2^{j-s}$. Bound the first s selected caps by c_j and sum. The cap differences telescope to $\sum_{j=L-s+1}^L c_j \leq sr$, proving (8). Now use $N_j \leq \min(2^j, R)$ and take expectations. Each term in the last sum has expectation $2^{j-s} 2^{-j} = 2^{-s}$, and $\mathbb{E}R = 1$. $\square$

Theorem 5 (Disproof of the uniform-constant conjecture). *For every real K there is a unit-coefficient multilinear polynomial on a unit cube, at a strictly interior point with positive hull gap, for which $T/H > K$. In particular, the Luedtke–Namazifar–Linderoth uniform-constant conjecture is false.*

Proof. Choose $s = \lceil \log_2 L \rceil$ in (9); then $s \leq L$ and $(L - s)2^{-s} \leq 1$. Hence

$$\frac{T_L}{H_L} \geq \frac{L}{\lceil \log_2 L \rceil + 1} \rightarrow \infty.$$

All coefficient, support, interiority, and positive-denominator conditions were checked in (5) and Lemma 3. A unit cube is an admissible nonnegative box. $\square$

The degree grows with L , so the theorem is consistent with finite degree-dependent bounds (Section 5).

4 The exact hull gap and an attaining law

The finite bound is attained at a suitable cutoff. Define

$$B_q = (L - q + 2)2^{-q} \quad (1 \leq q \leq L).$$

This sequence decreases strictly, with $B_1 \geq 1 \geq B_L$. There is therefore an integer $1 \leq s < L$ such that

$$B_{s+1} \leq 1 \leq B_s, \quad \text{or equivalently} \quad (L - s + 1)2^{-(s+1)} \leq 1 \leq (L - s + 2)2^{-s}. \quad (10)$$

Theorem 6 (Exact finite gap). *For every $L \geq 2$ and every cutoff s satisfying (10),*

$$T_L = L, \quad H_L = s + (L - s)2^{-s}. \quad (11)$$

Both envelope endpoints are attained. Either valid adjacent cutoff at an equality in (10) gives the same value.

Proof. The upper bound for H_L is Lemma 4. To attain it, first choose $l \in \{0, \dots, L\}$ with probabilities

$$w_l = \begin{cases} 2^{-l-1}, & l < L, \\ 2^{-L}, & l = L. \end{cases} \quad (12)$$

Set $A_j = \mathbf{1}\{j \leq l\}$. Summing the geometric tail gives $\mathbb{P}(A_j = 1) = 2^{-j}$. For a cutoff $q \in \{s, s+1\}$, prescribe

$$R_l^{(q)} = \begin{cases} 0, & l < q, \\ 2^{l-q+1}, & l \geq q. \end{cases} \quad (13)$$

The failure count is at most 2^L , since $q \geq 1$. A direct sum gives

$$\mathbb{E}R^{(q)} = (L - q)2^{-q} + 2^{1-q} = B_q.$$

Independently of l , mix the two choices of q with weights

$$\theta = \frac{1 - B_{s+1}}{B_s - B_{s+1}}, \quad 1 - \theta = \frac{B_s - 1}{B_s - B_{s+1}}. \quad (14)$$

They lie in $[0, 1]$ and make the mixed failure count have mean one.

We next realize the block hits and each individual leaf mean. If the chosen failure count is zero, let every leaf succeed. If it is $R = 2^r$, choose a uniform residue t modulo 2^{L-r} and fail exactly the leaves whose indices are congruent to t modulo 2^{L-r} . This gives 2^r failed leaves and conditional failure probability $R/2^L$ for every leaf. At level j , a block has length 2^{L-j} . If $j \leq r$, each block contains a failure, so the number hit is 2^j . If $j > r$, the spacing between failures exceeds the block length, and the 2^r failures hit different blocks. In both cases,

$$N_j = \min(2^j, R) \quad (15)$$

simultaneously for all levels. Averaging $R/2^L$ over the mixed law gives failure probability 2^{-L} for each leaf. Thus the entire law has the original prescribed means.

It remains to check the attained objective. For the nested anchors put $S_l(R) = \sum_{j=1}^l \min(2^j, R)$. For $l > s$ and either $R = 2^{l-s}$ or $R = 2^{l-s+1}$,

$$S_l(R) = sR + 2^{l-s+1} - 2 = sR + \sum_{j=s+1}^l 2^{j-s}.$$

Indeed, cap the first $l - s$ terms at 2^j and the last s at R ; the two endpoint choices both give equality. These are exactly the failure counts in (13). For $l = s$ they are 0 and 2, both satisfying $S_s(R) = sR$; for $l < s$ they are both zero. Therefore (8) is an equality in every state of the mixed law. Taking expectations and using (15) gives $\mathbb{E} \sum_j A_j N_j = s + (L - s)2^{-s}$. This attains the maximum in (7), and hence attains the lower envelope $L - H_L$. The common upper law attains L .

Finally, the formulas at s and $s + 1$ differ by $B_{s+1} - 1$, so they agree whenever both adjacent choices are valid. $\square$

L	A valid s	H_L	T_L/H_L
2	1	3/2	4/3
8	2	7/2	16/7
16	3	37/8	128/37
64	5	219/32	2048/219

Table 1: Exact instances of Theorem 6.

The cutoff also explains the growth. From $2^s \leq L - s + 2 \leq L + 1$, $s \leq \log_2(L + 1)$. The other cutoff inequality gives $2^{s+1} \geq L - s + 1$, which, together with this upper bound, yields $s = \log_2 L + O(1)$. Moreover,

$$0 \leq (L - s)2^{-s} \leq 2 \frac{L - s}{L - s + 1} < 2.$$

Consequently,

$$H_L = \log_2 L + O(1), \quad \frac{T_L}{H_L} \sim \frac{L}{\log_2 L}. \quad (16)$$

Nesting of the partitions was used for simultaneous attainment in (15). The finite bound and disproof themselves only need the specified equal block sizes and disjointness within each level.

5 The sharp replacement

Here and below $\ln$ denotes the natural logarithm.

Theorem 7 (Sharp degree and dimension growth). *For the suprema defined in Section 1,*

$$R(d) \sim \frac{\ln d}{\ln \ln d}, \quad C(n) \sim \frac{\ln n}{\ln \ln n}. \quad (17)$$

Both limits hold along all positive integer allowances tending to infinity, with leading constant one, over all finite nonnegative boxes.

The upper half rests on the following explicit bound, which is the finite harmonic certificate verified in Lean.

Theorem 8 (A common-coupling finite bound). *Let $d \geq 2$, $c_0 = 1 - e^{-1}$, and define*

$$\Lambda = \max\{e^6, 1 + \ln d\}, \quad b = \ln \Lambda, \quad A = \Lambda/b^2, \quad (18)$$

$$h = \frac{1 - 1/b}{1 + 1/b^2} \frac{b - 3 \ln b}{\Lambda}, \quad Z = \frac{1}{h} + A + \frac{2}{c_0}. \quad (19)$$

Then $h > 0$ and $T_B(f; x) \leq ZH_B(f; x)$ for every positive multilinear polynomial of degree at most d on a finite nonnegative box, including points of zero hull gap. On a unit cube the bound is witnessed by a single law depending only on the complete marginal vector and d , independently of the supports and coefficients.

Appendix A proves this theorem. The constant Z is chosen to give the sharp leading term; we do not claim it is optimal at any fixed degree.

Proof of Theorem 7. For the upper bound, $b \rightarrow \infty$ and

$$\frac{h}{b/\Lambda} = \frac{1 - 1/b}{1 + 1/b^2} \left(1 - \frac{3 \ln b}{b}\right) \rightarrow 1.$$

Thus $1/h \sim \Lambda/\ln \Lambda$, while $A/(\Lambda/\ln \Lambda) = 1/b \rightarrow 0$ and the constant $2/c_0$ is negligible. Eventually $\Lambda = 1 + \ln d$, so Theorem 8 gives

$$R(d) \leq (1 + o(1)) \frac{\ln d}{\ln \ln d}.$$

Every support in n coordinates has size at most n , so $C(n) \leq R(n)$.

For a common lower argument, take an integer $N \geq 8$ and $L = \lfloor \log_2 N \rfloor - 1$. Then $L \geq 2$ and $n_L = 2^L + L \leq 2^{L+1} \leq N$. The family therefore fits degree allowance N . It also fits exactly N coordinates after adding unused variables with arbitrary fixed cube means, for instance $1/2$. Projection of any law on the added coordinates gives an old law, and any old law extends by independent added coordinates. Both gaps are therefore unchanged. By (16), both $R(N)$ and $C(N)$ are at least

$$\frac{T_L}{H_L} \sim \frac{L}{\log_2 L} \sim \frac{\ln N}{\ln \ln N}.$$

This construction works for every $N \geq 8$. Together with the upper bounds it proves both limits. In particular, the relevant ratio sets are nonempty and bounded above before taking their suprema. $\square$

The lower family is sparse: it has fewer than $2n_L$ terms and $O(n_L \log n_L)$ variable occurrences.

A The common-coupling upper bound

This appendix proves Theorem 8. The parameters and the rational union estimate follow the formal proof.

A.1 Deficiencies and the easy terms

Discard constant and affine terms, which have zero width and do not change H . For a remaining support e , choose an anchor i_e of minimum mean and write

$$u_e = x_{i_e}, \quad p_j = 1 - x_j \ (j \neq i_e), \quad t_e = \min \left(u_e, \sum_{j \in e \setminus \{i_e\}} p_j \right), \quad D_e(X) = X_{i_e} - \prod_{i \in e} X_i. \quad (20)$$

For binary X , D_e is the nonnegative indicator that the anchor succeeds and at least one other coordinate fails. Lemmas 1 and 2 imply

$$T(f; x) = \sum_e a_e t_e, \quad H(f; x) = \max_{P \in \mathcal{P}(x)} \sum_e a_e \mathbb{E}_P D_e. \quad (21)$$

Thus a law satisfying $\mathbb{E}D_e \geq t_e/Z$ for every support bounds the ratio after summing with nonnegative coefficients. The law must be chosen once for the entire vector x .

Call a coordinate low if $x_i \leq 1/2$ and high otherwise. Independent Bernoulli rounding gives

$$\mathbb{E}D_e = u_e \left(1 - \prod_{j \in e \setminus \{i_e\}} (1 - p_j) \right).$$

If there are at least two low coordinates in the support, a nonanchor has $p_j \geq 1/2$, so the deficiency is at least $u_e/2 \geq t_e/2$. If there are no low coordinates, let $S = \sum_j p_j$. The elementary inequality $1 - p \leq e^{-p}$ gives

$$\mathbb{E}D_e \geq u_e(1 - e^{-S}) \geq c_0 u_e \min(1, S) \geq c_0 t_e/2.$$

For the middle inequality, use concavity of $1 - e^{-S}$ on $[0, 1]$ and monotonicity after 1. For the last, if $S \leq 1$ use $u_e > 1/2$; otherwise use $t_e \leq u_e$. Independence therefore captures at least $c_0 t_e/2$ in these two cases. Terms of zero gap need no division and satisfy the desired inequality under every law.

A.2 Two laws for the one-low case

Put $M = e^{\Lambda-1} \geq d$, so $1 + \ln M = \Lambda$. Each of the next two laws draws one uniform $U \in (0, 1)$ and sets every low coordinate to $X_i = \mathbf{1}\{U \leq x_i\}$.

In the threshold law, a high coordinate with failure mean $p = 1 - x_i$ fails exactly when $U \leq p$. All means are preserved. For a one-low support with low-anchor mean u , this law gives deficiency

$$\min \left(u, \max_j p_j \right). \quad (22)$$

In the harmonic law, a high coordinate with $p = 0$ never fails. For $p > 0$, put

$$h_p = \min(Mp, 1), \quad a_p = 1 + \ln(h_p/p), \quad q_p(t) = \frac{\min(1, p/t)}{a_p} \mathbf{1}\{t \leq h_p\} \quad (0 < t < 1). \quad (23)$$

Conditional on $U = t$, high coordinates fail independently with probabilities $q_p(t)$. Since $p \leq h_p$ and $1 \leq a_p \leq \Lambda$, these probabilities belong to $[0, 1]$. They are measurable and

$$\int_0^1 q_p(t) dt = \frac{p + p \ln(h_p/p)}{a_p} = p. \quad (24)$$

Hence every high coordinate also has the prescribed mean. Values at the endpoints of the integral have no effect. The finite binary law can equivalently be defined by integrating the conditional probability of each binary vertex; finite sums commute with the integral.

For a one-low support its anchor is the low coordinate. Conditional on $U = t \leq u$, the probability of a nonanchor failure is $1 - \prod_j (1 - q_{p_j}(t))$. Its harmonic deficiency is therefore

$$\int_0^u \left[1 - \prod_j (1 - q_{p_j}(t)) \right] dt. \quad (25)$$

Both laws are defined using the complete marginal vector, without reference to a support or its coefficient.

A.3 The harmonic window and the mixture

First, for independent events of probabilities $q_j \in [0, 1]$ and $S = \sum_j q_j$,

$$1 - \prod_j (1 - q_j) \geq \frac{S}{1 + S}. \quad (26)$$

Indeed, $\prod_j (1 + q_j) \geq 1 + S$ and $\prod_j (1 - q_j) \prod_j (1 + q_j) = \prod_j (1 - q_j^2) \leq 1$. Multiply the first inequality by the nonnegative failure-complement product and rearrange. The right side is increasing in $S \geq 0$.

Let $\beta = 1/b$, with b, A, h as in (18)–(19). We have $b \geq 6$, and $b - 3 \ln b > 0$: it is positive at 6 (since $e^2 > 6$) and its derivative is $1 - 3/b > 0$ thereafter. Thus $h > 0$ and

$$\ln(\beta A) = b - 3 \ln b > 0, \quad \beta A > 1, \quad A > 1.$$

For a one-low term with $t_e > 0$, if some $p_j \geq t_e/A$, then (22) is at least t_e/A , since $t_e \leq u$. Otherwise every $p_j < t_e/A$. Consider the nonempty window

$$t_e/A \leq t \leq \beta t_e \leq u. \quad (27)$$

All $p_j \leq t$ there. Call a coordinate active when $t \leq M p_j$. Since $t \leq 1$, this is the same as $t \leq h_{p_j}$. Every inactive coordinate has $p_j < t/M$, and there are at most $d - 1 \leq M$ coordinates in the nonanchor support. The total inactive failure mass is at most t , so the active mass is at least

$$\sum_j p_j - t \geq t_e - t \geq (1 - \beta)t_e.$$

On active coordinates, (23) gives $q_{p_j}(t) \geq p_j/(\Lambda t)$. Consequently,

$$\sum_j q_{p_j}(t) \geq z(t) := \frac{(1 - \beta)t_e}{\Lambda t}, \quad 0 \leq z(t) \leq \frac{A}{\Lambda} = \frac{1}{b^2}.$$

Using (26), the conditional union probability is at least $z(t)/(1 + z(t)) \geq z(t)/(1 + 1/b^2)$. Integrate over (27), using nonnegativity outside that window, to obtain

$$\begin{aligned} \mathbb{E}_{\text{harm}} D_e &\geq \frac{1 - \beta}{1 + 1/b^2} \frac{t_e}{\Lambda} \ln(\beta A) \\ &= \frac{1 - 1/b}{1 + 1/b^2} \frac{b - 3 \ln b}{\Lambda} t_e = h t_e. \end{aligned}$$

The integrands are bounded measurable functions on a finite interval, and the reciprocal integration window stays strictly above zero.

Mix the harmonic, threshold, and independent laws with weights

$$\frac{1/h}{Z}, \quad \frac{A}{Z}, \quad \frac{2/c_0}{Z}, \quad Z = 1/h + A + 2/c_0. \quad (28)$$

These positive weights sum to one and preserve every coordinate mean. Each positive-gap term receives at least t_e/Z from one of the three components. All other components contribute nonnegative deficiencies. Summing (21) proves $T \leq ZH$ on the cube, including at zero hull gap. This proves the cube part of Theorem 8.

A.4 Transfer to original terms on a nonnegative box

For each coordinate substitute $x_i = \ell_i + (u_i - \ell_i)y_i$ with $0 \leq y_i \leq 1$, and let $\tilde{f}(y) = f(x)$. Expanding original monomials gives nonnegative coefficients and does not increase degree.

First suppose all coordinates are nonfixed. The substitution is an affine bijection, so it maps graph hulls and preserves their vertical widths:

$$H_B(f; x) = H(\tilde{f}; y), \quad T_B(f; x) \leq T(\tilde{f}; y). \quad (29)$$

For the second inequality, an original term becomes a sum of expanded monomials. The sum of their convex envelopes is a convex underestimator of that term; the sum of their concave envelopes is a concave overestimator. Thus the original term's width is at most the sum of the expanded widths. Sum this inequality over the original terms. If expansion produces the same support more than once, combining its nonnegative coefficients leaves that sum of widths unchanged, because envelopes scale linearly by a nonnegative constant.

If a coordinate is fixed, restrict the original box to its nonfixed coordinates. In the expanded function the fixed coordinate has zero width and its new cube coordinate is unused. Projecting or extending laws on unused coordinates preserves every attainable expectation, as in the proof of Theorem 7. Thus both the full-hull equality and the original-term inequality in (29) still hold. If all coordinates are fixed, every gap is zero. Applying the cube bound to $\tilde{f}$ now proves $T_B(f; x) \leq T(\tilde{f}; y) \leq ZH(\tilde{f}; y) = ZH_B(f; x)$, completing Theorem 8.

B Formal statements and reproduction

The [formal/](#) directory is a standalone Lake project pinned to Lean 4.33.1 and Mathlib v4.33.1, with exact dependency revisions in `lake-manifest.json`; it needs no other repository files. Its 48 proof modules are 41 multilinear modules and seven shared modules for finite laws, graph hulls, and individual envelopes. The shared modules keep the namespace `CubicGap` from their original location; they are general infrastructure, and no cubic counterexample modules are included.

The file links below are relative to this PDF; [formal/COVERAGE.md](#) lists the same paths and declaration names for viewers that block local links.

Paper claim	Lean file and principal declarations
Lemma 1, continuous cube hull	Hull.lean : <code>mem_cubeGraph_hull_iff</code> .
Lemma 3, $T_L = L$ and $H_L > 0$	Termwise.lean : <code>termwiseGap_eq</code> ; Results.lean : <code>hullGap_positive</code> .
Lemma 4	ExactUpper.lean : <code>dyadic_exact_pointwise</code> , <code>hullGap_exact_upper_bound</code> .
Theorem 5	Results.lean : <code>unbounded_gap_ratio</code> , <code>no_uniform_positive_multilinear_bound</code> .
Theorem 6	ExactResults.lean : <code>polynomial_exact_minimum</code> , <code>hullGap_exact</code> , <code>exists_hullGap_exact</code> .
Theorem 8, cube	SharpUpper.lean : <code>sharpLaw_hasMeans</code> , <code>sharp_cube_gap_bound</code> .
Theorem 8, original boxes	BoxTransfer.lean : <code>degree_gap_bound_on_box</code> .
Theorem 7	SharpAsymptotics.lean : <code>sharp_positive_growth</code> .

All multilinear declarations in the table belong to namespace `MultilinearGap`, and the shared hull declarations to `CubicGap`. The final asymptotic theorem states the two normalized limits for the actual all-box degree and exactly- n -coordinate suprema, not merely along the dyadic family.

The formal disproof and asymptotic lower bound use the weaker verified estimates $H_L \leq s+2L/2^s$ and $H_L \leq \log_2 L + 3$, not the exact formula or attaining law; the written disproof uses the sharper Lemma 4, which is also verified.

Supporting claims. General endpoint attainment is proved in `Attainment.lean`. The module `EnvelopeFunctions.lean` identifies those endpoints with the greatest convex underestimator and least concave overestimator on both cubes and boxes. `MonomialEnvelope.lean` proves the full individual lower-envelope identity in Lemma 2, including an attaining law preserving all ambient means. `PaperFoundations.lean` checks original-box coefficient scaling and $0 \leq H_B \leq T_B$. `ExactSize.lean` proves all four size identities in (6), with the coordinate count inherited from `FamilySize.lean`, and the stated sparsity bounds. `ExactAsymptotics.lean` proves $|H_L - \log_2 L| \leq 3$ for $L \geq 2$ and the actual family ratio limit in (16). `Examples.lean` proves all four rows of Table 1 as equalities about the actual graph hull.

Reproduction. With Elan, Git, and Python 3 installed and the pinned toolchain on `PATH`, run from `formal/`:

```
lake exe cache get
bash scripts/verify.sh
```

The script checks import coverage, builds with warnings as failures, audits the transitive axiom dependencies of all bundled declarations, and replays the project declarations through Lean’s kernel. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted, so unfinished proofs and other axioms are rejected; no solver or sampled numerical result is a trusted step.

Trust boundary and scope. The replay uses the installed Lean kernel and compiled dependencies; it is not a separate kernel implementation and does not replay all of Mathlib from scratch. The mathematical interpretation also depends on matching the formal definitions to the statements above. The [verification record](#) specifies the actual local checks and their scope. The formal scope is the disproof, the exact family gap and attainment, the finite bound (19), and the sharp leading limits. Optimized fixed-point or Lambert W formulas, second-order asymptotics, coefficient removal, and homogenization are outside this paper; no theorem here needs them.

References

- [1] Natasha Boland, Santanu S. Dey, Thomas Kalinowski, Marco Molinaro, and Fabian Rigterink. Bounding the gap between the McCormick relaxation and the convex hull for bilinear functions. *Mathematical Programming*, 162(1–2):523–535, 2017. doi: 10.1007/s10107-016-1031-5.
- [2] James Luedtke, Mahdi Namazifar, and Jeff Linderoth. Some results on the strength of relaxations of multilinear functions. *Mathematical Programming*, 136(2):325–351, 2012. doi: 10.1007/s10107-012-0606-z. Conjecture 4.1, p. 349; Conjecture 1, p. 22 in the author manuscript, available at <https://jlinderoth.github.io/papers/Luedtke-Namazifar-Linderoth-12-TR.pdf>.